the eigenvalues of $\operatorname{Hess} \mathsf{E}_0(\overline{X})$ are non-positive. Per Theorem 4.1, this implies that $x_1 = \ldots = x_n$. But then, for large enough k, X_k is also constituted of points which are all nearly equal, whence in the same hemisphere, and the only such critical point of E_β is that in which all points are equal (synchronized). This, combined with the continuity of the eigenvalues of $\operatorname{Hess} \mathsf{E}_\beta$ with respect to β and (C.2), proves that there exists some $c > 0$ independent of n such that whenever $\beta \leqslant c\, n^{-1}$, all critical points of E_β except synchronized ones are strict saddle points. □

Remark C.1. *We comment on the extension of the above proof to the dynamics* (SA)*. The point of contention is* (C.1)*, since the metric with respect to which the gradient and Hessian of* E_0 *are taken is not the same as that for* E_β*. Denote the modified metric defined in Section 3.4 by* g_β*, and the canonical metric by* g*. For any* $x \in (\mathbb{S}^{d-1})^n$ *and* $v \in \mathrm{T}_x(\mathbb{S}^{d-1})^n$ *we have*

$$\mathrm{D}\mathsf{E}_\beta(x)[v] = g_\beta(\nabla_{g_\beta} \mathsf{E}_\beta(x), v),$$

but also $\mathrm{D}\mathsf{E}_\beta(x)[v] = g(\nabla_g \mathsf{E}_\beta(x), v)$*. By virtue of the explicit form of* g_β *and* E_β *as well as* (C.1)*, we gather that*

$$g_\beta(\nabla_{g_\beta} \mathsf{E}_\beta(x), v) = g(\nabla_g \mathsf{E}_0(x), v) + O(\beta) \tag{C.3}$$

which implies that any sequence of critical points of E_β *converges to a critical point for* E_0*. Similarly, since* $\operatorname{Hess}_{g_\beta} \mathsf{E}_\beta(x)[v] = \mathrm{D}(\nabla_{g_\beta} \mathsf{E}_\beta(x))[v]$*, we find*

$$\operatorname{Hess}_{g_\beta} \mathsf{E}_\beta(x)[v] = \operatorname{Hess}_g \mathsf{E}_0(x)[v] + O(\beta). \tag{C.4}$$

We can then repeat the argument in the proof above by replacing (C.1) *by* (C.3) *and* (C.4)*.*

Appendix D. Proof of Theorem 6.9

Proof. We focus on the dynamics (SA), since the proof for (USA) follows from very similar computations.

Step 1. The flow map is Lipschitz. We begin by showing that the trajectories satisfy a Lipschitz property with respect to the initial data. To this end, let $(x_i(\cdot))_{i\in[n]} \in C^0(\mathbb{R}_{\geqslant 0}; (\mathbb{S}^{d-1})^n)$ and $(y_i(\cdot))_{i\in[n]} \in C^0(\mathbb{R}_{\geqslant 0}; (\mathbb{S}^{d-1})^n)$ be two solutions to the Cauchy problem for (SA) associated to data $(x_i(0))_{i\in[n]}$ and $(y_i(0))_{i\in[n]}$ respectively. For any $i \in [n]$ and $t \geqslant 0$, we have

$$\begin{aligned} x_i(t) - y_i(t) &= x_i(0) - y_i(0) \\ &\quad + \int_0^t \sum_{j=1}^n \left(\frac{e^{\beta\langle x_i(s), x_j(s)\rangle}}{\sum_{k=1}^n e^{\beta\langle x_i(s), x_k(s)\rangle}} \right) \big(x_j(s) - \langle x_i(s), x_j(s)\rangle x_i(s) \big)\, \mathrm{d}s \\ &\quad - \int_0^t \sum_{j=1}^n \left(\frac{e^{\beta\langle y_i(s), y_j(s)\rangle}}{\sum_{k=1}^n e^{\beta\langle y_i(s), y_k(s)\rangle}} \right) \big(y_j(s) - \langle y_i(s), y_j(s)\rangle y_i(s) \big)\, \mathrm{d}s. \end{aligned} \tag{D.1}$$

We see that

$$\left\| \int_0^t \sum_{j=1}^n \left(\frac{e^{\beta\langle x_i(s), x_j(s)\rangle}}{\sum_{k=1}^n e^{\beta\langle x_i(s), x_k(s)\rangle}} \right) (x_j(s) - y_j(s))\, \mathrm{d}s \right\| \leqslant \int_0^t \max_{j\in[n]} \|x_j(s) - y_j(s)\|\, \mathrm{d}s. \tag{D.2}$$

On another hand, since the softmax function with a parameter β is β–Lipschitz (with respect to the Euclidean norm), we also get

$$\begin{aligned}
&\left\| \int_0^t \sum_{j=1}^n \left(\frac{e^{\beta\langle x_i(s), x_j(s)\rangle}}{\sum_{k=1}^n e^{\beta\langle x_i(s), x_k(s)\rangle}} - \frac{e^{\beta\langle y_i(s), y_j(s)\rangle}}{\sum_{k=1}^n e^{\beta\langle y_i(s), y_k(s)\rangle}} \right) y_j(s)\, \mathrm{d}s \right\| \\
&\qquad \leqslant \beta n^{\frac{1}{2}} \int_0^t \left(\sum_{j=1}^n \Big[\langle x_i(s), x_j(s)\rangle - \langle y_i(s), y_j(s)\rangle \Big]^2 \right)^{\frac{1}{2}} \mathrm{d}s \\
&\qquad \leqslant 2\beta n \int_0^t \max_{j\in[n]} \|x_j(s) - y_j(s)\|\, \mathrm{d}s. \qquad \text{(D.3)}
\end{aligned}$$

Using (D.2), (D.3) and arguing similarly for the remaining terms in (D.1), we deduce that

$$\|x_i(t) - y_i(t)\| \leqslant \|x_i(0) - y_i(0)\| + 10 \max\{1, \beta\} n \int_0^t \max_{j\in[n]} \|x_j(s) - y_j(s)\|\, \mathrm{d}s.$$

Maximizing over $i \in [n]$ and applying the Grönwall inequality yields

$$\max_{j\in[n]} \|x_j(t) - y_j(t)\| \leqslant c(\beta)^{nt} \max_{j\in[n]} \|x_j(0) - y_j(0)\|, \qquad \text{(D.4)}$$

for any $i \in [n]$ and $t \geqslant 0$.

Step 2. Almost orthogonality. Let $x_1(0), \dots, x_n(0) \in \mathbb{S}^{d-1}$ be the random i.i.d. initial points. We prove that with high probability, there exist n pairwise orthogonal points $y_1(0), \dots, y_n(0) \in \mathbb{S}^{d-1}$, such that for any $i \in [n]$,

$$\|x_i(0) - y_i(0)\| \leqslant \sqrt{\frac{\log d}{d}}. \qquad \text{(D.5)}$$

To this end, we take $y_1(0) = x_1(0)$ and then construct the other points $y_i(0)$ by induction. Assume that $y_1(0), \dots, y_i(0)$ are constructed for some $i \in [n]$, using only knowledge about the points $x_1(0), \dots, x_i(0)$. Then by Lévy's concentration of measure, since $x_{i+1}(0)$ is independent from $x_1(0), \dots, x_i(0)$ and uniformly distributed on $\mathbb{S}^{d-1}$,

$$\mathbb{P}\left(\left\{ \operatorname{dist}\left(x_{i+1}(0), \operatorname{span}\{y_1(0), \dots, y_i(0)\}^\perp\right) \leqslant \sqrt{\frac{\log d}{d}} \right\} \right) \geqslant 1 - 4id^{-1/64},$$

for some universal constants $c, C > 0$. Using the union bound, we gather that the event

$$\mathscr{A}_0 = \{\text{(D.5) is satisfied for any } i \in [n]\}$$

has probability at least $p_0 = 1 - 2n^2 d^{-1/64}$. We now consider the event

$$\mathscr{A} = \mathscr{A}_0 \cap \{\text{for some } C, \lambda > 0, \text{ (6.1) holds for any } i \in [n] \text{ and } t \geqslant 0\}$$

which, since $d \geqslant n$ and thus the second event has probability 1, also holds with probability at least $p_0 = 1 - 2n^2 d^{-1/64}$. For the remainder of the proof, we assume that $\mathscr{A}$ is satisfied.

Step 3. Proof of (6.11). Let $(y_i(\cdot))_{i\in[n]} \in C^0(\mathbb{R}_{\geqslant 0}; (\mathbb{S}^{d-1})^n)$ denote the unique solution to the Cauchy problem for (SA) corresponding to the initial datum $(y_i(0))_{i\in[n]}$. A combination of (D.4) and (D.5) yields

$$\|x_i(t) - y_i(t)\| \leqslant c(\beta)^{nt}\sqrt{\frac{\log d}{d}} \tag{D.6}$$

for any $i \in [n]$ and $t \geqslant 0$, under $\mathscr{A}$. Combining (D.6) with Theorem 6.8 we obtain

$$\Big|\langle x_i(t), x_j(t)\rangle - \gamma_\beta(t)\Big| \leqslant 2c(\beta)^{nt}\sqrt{\frac{\log d}{d}} \tag{D.7}$$

for any $i \neq j$ and $t \geqslant 0$, under $\mathscr{A}$.

We turn to the proof of the second part of (6.11). For this, we prove that for large times t, both $\gamma_\beta(t)$ and $\langle x_i(t), x_j(t)\rangle$ are necessarily close to 1. We first show that

$$1 - \gamma_\beta(t) \leqslant \frac{1}{2}\exp\left(\frac{n^2 e^\beta}{2\left(n + e^{\frac{\beta}{2}}\right)} - \frac{nt}{n + e^{\frac{\beta}{2}}}\right) \tag{D.8}$$

for any $t \geqslant 0$. To this end, we notice that $t \mapsto \gamma_\beta(t)$ is increasing and thus $\gamma_\beta(t) \geqslant 0$, as well as $\dot\gamma_\beta(t) \geqslant \frac{1}{ne^\beta}$ as long as $\gamma_\beta(t) \leqslant \frac{1}{2}$. Therefore,

$$\gamma_\beta\left(\frac{ne^\beta}{2}\right) \geqslant \frac{1}{2}.$$

We deduce that for $t \geqslant \frac{ne^\beta}{2}$,

$$\dot\gamma_\beta(t) \geqslant \frac{n(1-\gamma_\beta(t))}{n + e^{\frac{\beta}{2}}}.$$

Integrating this inequality from $\frac{ne^\beta}{2}$ to t, we obtain (D.8). We now set $d^*(n,\beta) \geqslant n$ such that

$$\frac{d}{\log d} \geqslant \frac{16c(\beta)^2}{\gamma_\beta(\frac{1}{n})^2} \tag{D.9}$$

holds for any $d \geqslant d^*(n,\beta)$. According to Lemma 6.4, since $\mathscr{A}$ is satisfied, there exists $x^* \in \mathbb{S}^{d-1}$ such that $x_i(t) \to x^*$ for any $i \in [n]$ as $t \to +\infty$. We set

$$\alpha(t) := \min_{i\in[n]}\langle x_i(t), x^*\rangle,$$

and prove that

$$1 - \alpha(t) \leqslant \exp\left(\frac{1 - \gamma_\beta\left(\frac{1}{n}\right)t}{2ne^{2\beta}}\right). \tag{D.10}$$

To this end, let us first prove that

$$\alpha\left(\frac{1}{n}\right) \geqslant \frac{1}{2}\gamma_\beta\left(\frac{1}{n}\right). \tag{D.11}$$

From Step 2 in the proof of Lemma 6.4, we gather that x^* lies in the convex cone generated by the points $x_1(t), \dots, x_n(t)$ for any $t > 0$, and so the decomposition